Computing Basics -- Guided Notes ANSWER KEY

Unit 1.2 | CompTIA Network+ N10-009 Obj. Core 1 1.1 | N10-008

For instructor use / distribute after students complete the notes.

Question 1

Every computer performs four basic functions: _______ (keyboard, mouse), _______ (CPU), _______ (RAM, hard drive), and _______ (monitor, printer). The _______ (CPU) is considered the brain of the computer because it coordinates all four functions.

Answer: Input; processing; storage; output; CPU (Central Processing Unit).

Question 2

The CPU's fetch-decode-execute cycle has three steps: it _______ an instruction from memory, _______ the instruction to determine what operation to perform, and then _______ the operation. The speed at which this cycle repeats is measured in _______, with modern CPUs reaching billions of cycles per second (gigahertz).

Answer: Fetches; decodes; executes; clock speed (Hz / GHz).

Question 3

A modern CPU has multiple _______, each capable of executing instructions independently. Hyper-threading (Intel) and SMT (AMD) allow each core to handle two _______ simultaneously. A 6-core CPU with hyper-threading appears to the operating system as _______ logical processors.

Answer: Cores; threads; 12 logical processors.

Question 4

CPU cache is ultra-fast memory built directly into the processor. _______ cache is the smallest and fastest, holding recently used data for the specific core. _______ cache is slightly larger and shared within a core. _______ cache is the largest and is shared across all cores. Cache reduces the need to access _______, which is much slower.

Answer: L1 cache; L2 cache; L3 cache; RAM (main memory).

Question 5

RAM (Random Access Memory) is described as _______ memory because its contents are lost when power is removed. A typical modern desktop uses _______ RAM. The data currently in RAM represents the programs and files that are _______ open and in use.

Answer: Volatile; DDR4 or DDR5; currently (actively).

Question 6

When the system runs low on RAM, the OS uses a _______ file (also called virtual memory) on the hard drive as overflow space. This is stored as _______ on Windows. Because hard drive access is much slower than RAM, heavy use of virtual memory results in noticeable system _______.

Answer: Page file; pagefile.sys; slowdown (sluggishness / poor performance).

Question 7

A traditional mechanical hard drive (HDD) has typical sequential read speeds of _______ MB/s, while a modern NVMe SSD can reach _______ MB/s or higher. The PSU provides three main DC voltage rails: _______ V (for CPU and GPU), _______ V, and _______ V.

Answer: 80-160 MB/s (HDD); 3000-7000 MB/s (NVMe); +12V; +5V; +3.3V.

Question 8 -- Real World Application

REAL WORLD: A user has a 3-year-old desktop with 8 GB of RAM and a mechanical hard drive. They recently upgraded to Windows 11 and opened their usual applications -- a browser with 20 tabs, a video editor, and a spreadsheet. The system is now extremely slow. Using the concepts of RAM, virtual memory, and storage performance, explain what is most likely happening and what upgrade would have the greatest impact on performance.

Answer: With 8 GB of RAM and 20 browser tabs plus a video editor and spreadsheet, the system is exceeding available RAM and is heavily using the page file on the mechanical hard drive. HDD random-access speeds (less than 150 MB/s) are far slower than RAM, causing severe performance degradation whenever virtual memory is used. The single most impactful upgrade would be replacing the HDD with an SSD, which reduces page file latency dramatically. A secondary upgrade would be adding more RAM (16 GB or 32 GB) to reduce page file usage entirely.